UiT The Arctic University of Norway



[Missin/teams name]

Participants	[your name]
Issue	[version of the report]
Date of Issue	[YYYY-MM-DD]
Course	STE-3609 System Engineering

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1

Introduction and Mission Definition

Provide a brief introduction to the project so that an external reader can understand the mission and its scope.

The chapter should include:

- A brief description of the Lunar Rover Delivery and Deployment Mission.
- The main mission objective and success criteria.
- The initial mission conditions.
- The scope and system boundary.
- The main assumptions and constraints.
- A brief background or literature review relevant to the proposed solution.

This chapter should be short and precise. Relevant information from the provided project description may be reused or adapted; it is not necessary to reproduce the complete project description.

2

Concept of Operations

Provide a brief and visual description of how the proposed mission concept operates, from the initial conditions to successful mission completion.

A mission timeline, operational schematic, or similar figure is recommended to show the main mission phases and their sequence. Include a short explanation of the most important activities and events.

Note: The mission concept presented during Week 1 provides the starting point for this chapter. You may reuse and refine those figures and descriptions, updating them where necessary to reflect the final mission concept.

Keep this chapter **short, visual, and focused**. The detailed MBSE and Capella analysis should be presented in the following chapter.

3

System Architecture Development

Present the development of the system architecture using ARCADIA and Capella:

Operational Analysis → System Analysis → Logical Architecture → Physical Architecture

Use selected Capella diagrams to explain the main architectural decisions and how the system evolved between the different analysis levels. Include only the diagrams necessary to understand and justify the proposed architecture.

Each selected diagram should be accompanied by a brief explanation. When appropriate, relate important architectural decisions to requirements, scenarios, calculations, simulations, or other engineering analyses.

Note: The complete Capella model is part of the project deliverables. Additional information may therefore be referenced using the corresponding Capella diagram name rather than reproduced in the report.

3.1 Operational Analysis

3.2 System Analysis

3.3 Logical Architecture

3.4 Physical Architecture

3.5 Requirements Integration

Briefly describe how requirements influenced the development of the system architecture and how relevant requirements are integrated into the Capella model.

It is not necessary to reproduce the requirements database. When a requirement is relevant to the discussion, refer to it directly using its Requirement ID (e.g., `REQ-005`). The complete requirements baseline is available in the submitted Excel workbook and Capella model.

4

Engineering Design and Analysis

Present the engineering analyses used to support and justify the proposed system design. These may include calculations, simulations, models, trade-off studies, or other relevant analyses.

Examples include:

- Trajectory, orbital, or landing analysis.
- Mass, power, or other system budgets.
- Propulsion, communication, ADCS, structural, thermal, or mechanical analyses.
- Simulations and numerical models.
- Trade-off studies between alternative solutions.

The analyses will depend on the solution developed by each team. Include only those relevant to demonstrating the feasibility of the proposed system.

Important: Engineering decisions should be justified using appropriate analysis. Explain why important solutions were selected and, where relevant, compare them with alternative approaches.

5

Verification and Project Maturity

The detailed verification and maturity information is maintained in the provided Excel workbook and does not need to be reproduced in the report.

Include the relevant summary graphs generated from the Excel workbook and briefly discuss what they indicate about the current state of the project.

The discussion should highlight the main observations regarding:

- Requirements and design maturity.
- Planned verification methods and levels.
- Areas requiring further development.
- Relevant gaps or concerns identified from the results.

Important: Do not include the graphs without interpretation. Provide your own assessment of what the results indicate about the current project maturity.

For detailed information, refer to the submitted Excel workbook.

/6

Conclusion

Summarize the main results and conclusions of the project, including an overall assessment of the feasibility and maturity of the proposed system.

Highlight the most important achievements, limitations, and areas requiring further development.

The conclusion should be short and concise and should not introduce new information.

Report length: As a general guideline, the main body of the report should be approximately **15–20 pages**. This is not a strict page requirement; shorter or longer reports are acceptable when justified by the project content.

The report should remain concise. Focus on the information necessary to understand and evaluate the proposed system, while detailed information should remain in the corresponding Capella model, Excel workbook, simulations, and other project artifacts.

7

List of Attachments

List all files submitted as attachments to this report. The purpose of this list is to allow the evaluators to verify that all referenced supporting material has been included in the submission.

- Attachment name
- Attachment name
- Attachment name

Bibliography example [1]

Bibliography

[1] G. D. Greenwade, "The Comprehensive Tex Archive Network (CTAN)," *TUGBoat*, vol. 14, no. 3, pp. 342–351, 1993.